



**PHYSICS
STANDARD LEVEL
PAPER 1A**

Topic A.2.3 Circular Motion and D.1 Gravitation Study Guide

May 2006 - May 2013

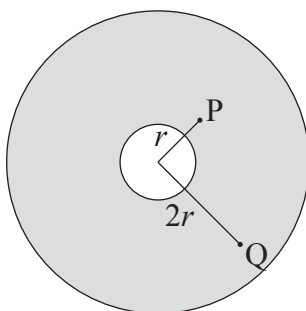
45 minutes

INSTRUCTIONS TO CANDIDATES

- Do not open this examination paper until instructed to do so.
- Answer all the questions.
- For each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.



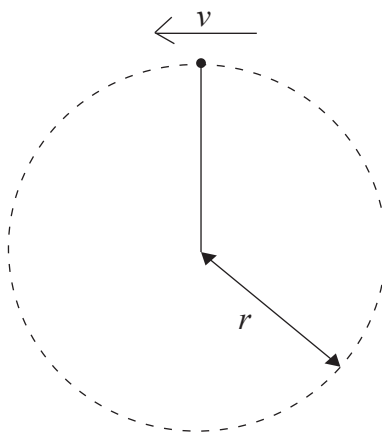
12. Two points P and Q are at distances r and $2r$ respectively from the centre of a compact disc (CD) as shown.



When the disc is rotating about its centre, the ratio of the $\frac{\text{acceleration at P}}{\text{acceleration at Q}}$ is

- A. $\frac{1}{2}$.
- B. 1.
- C. $\sqrt{2}$.
- D. 2.

13. A stone of mass m is attached to a string. The stone is made to rotate in a vertical circle of radius r , as shown.



At the point where the stone is vertically above the centre of the circle, the stone has speed v .

Which of the following expressions gives the tension in the string?

- A. $mg - \frac{mv^2}{r}$
- B. $\frac{mv^2}{r}$
- C. $\frac{mv^2}{r} - mg$
- D. $\frac{mv^2}{r} + mg$

8. A communications satellite is moving at a constant speed in a circular orbit around Earth. At any given instant in time, the resultant force on the satellite is
- A. zero.
 - B. equal to the gravitational force on the satellite.
 - C. equal to the vector sum of the gravitational force on the satellite and the centripetal force.
 - D. equal to the force exerted by the satellite's rockets.

19. The mass of Earth is M_E , its radius is R_E and the magnitude of the gravitational field strength at the surface of Earth is g . The universal gravitational constant is G . The ratio $\frac{g}{G}$ is equal to

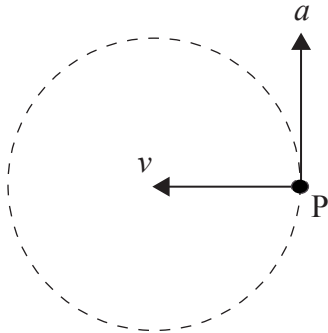
- A. $\frac{M_E}{R_E^2}$
- B. $\frac{R_E^2}{M_E}$
- C. $M_E R_E$
- D. 1

19. The gravitational force between two unit masses separated by a distance r is F_g . The electric force between two unit charges separated by a distance r is F_e . The Coulomb constant is k and the universal gravitational constant is G . The ratio F_g / F_e is
- A. one.
 - B. $\frac{k}{G}$.
 - C. $\frac{G}{k}$.
 - D. Gk .

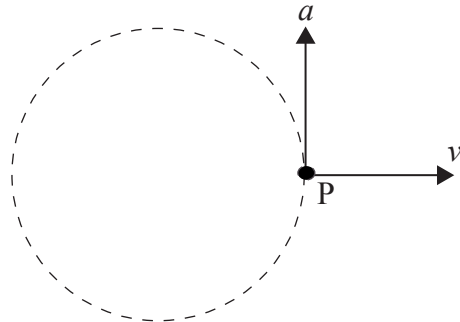
8. A particle P is moving anti-clockwise with constant speed in a horizontal circle.

Which diagram correctly shows the direction of the velocity v and acceleration a of the particle P in the position shown?

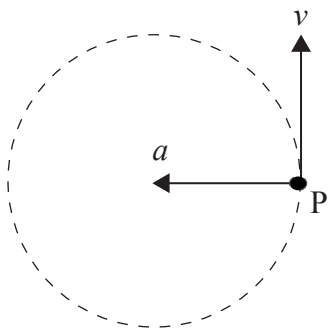
A.



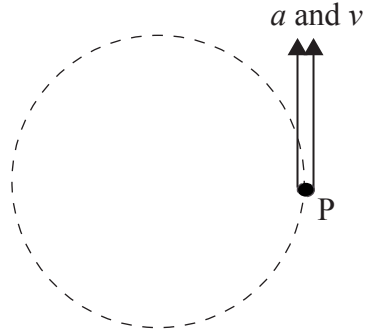
B.



C.



D.

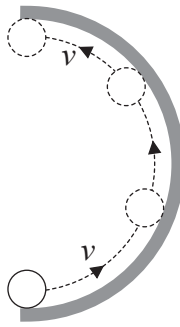


19. The weight of an object of mass 1 kg at the surface of Mars is about 4 N. The radius of Mars is about half the radius of Earth. Which of the following is the best estimate of the ratio below?

$$\frac{\text{mass of Mars}}{\text{mass of Earth}}$$

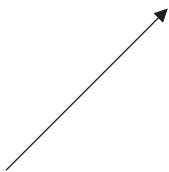
- A. 0.1
- B. 0.2
- C. 5
- D. 10

6. A ball moves along the inside of a horizontal semi-circular ring as shown. The diagram is a view from above.



Which arrow represents the direction of the average force on the ball?

A.



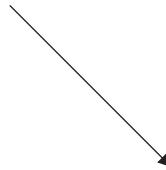
B.



C.



D.



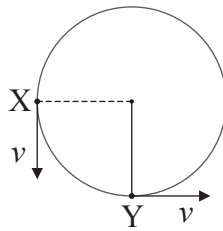
9. A cyclist rides around a circular track at a uniform speed. Which of the following correctly gives the net horizontal force on the cyclist at any given instant of time?

	Net horizontal force along direction of motion	Net horizontal force normal to direction of motion
A.	zero	zero
B.	zero	non zero
C.	non zero	zero
D.	non zero	non zero

20. A spherical planet of uniform density has three times the mass of the Earth and twice the average radius. The magnitude of the gravitational field strength at the surface of the Earth is g . What is the gravitational field strength at the surface of the planet?

- A. $6g$
- B. $\frac{2}{3}g$
- C. $\frac{3}{4}g$
- D. $\frac{3}{2}g$

7. A stone attached to a string is moving in a horizontal circle. The constant speed of the stone is v . The diagram below shows the stone in two different positions, X and Y.



Which of the following shows the direction of the change of velocity of the stone when moving from position X to position Y?

A.

B.

C.

D.

19. A spacecraft travels away from Earth in a straight line with its motors shut down. At one instant the speed of the spacecraft is 5.4 km s^{-1} . After a time of 600 s, the speed is 5.1 km s^{-1} . The average gravitational field strength acting on the spacecraft during this time interval is
- A. $5.0 \times 10^{-4} \text{ N kg}^{-1}$
 - B. $3.0 \times 10^{-2} \text{ N kg}^{-1}$
 - C. $5.0 \times 10^{-1} \text{ N kg}^{-1}$
 - D. 30 N kg^{-1}

8. A car moves at constant speed around a horizontal circular track. The resultant force on the car is always equal to
- A. the forward force from the engine.
 - B. the sideways friction between the tyres and the track.
 - C. the weight of the car.
 - D. zero.

19. A mass at point X gives rise to a gravitational field strength g at point P as shown below.



An identical mass is placed at point Y as shown below.



The resultant gravitational field strength at P is now

- A. greater than $2g$.
- B. between $2g$ and g .
- C. between g and zero.
- D. zero.

19. A particle of mass m is a distance R from the surface of Earth of mass M . The force acting on the particle is F . Which of the following is the gravitational field strength at R ?

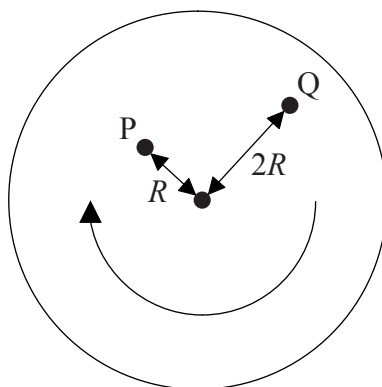
A. $\frac{Gm}{R^2}$

B. $\frac{GmM}{R^2}$

C. $\frac{F}{m}$

D. $\frac{F}{M}$

8. A horizontal disc is rotating about a vertical axis through its centre. Points P and Q on the disc are at distances R and $2R$ respectively from the centre.



The acceleration at P is a . Which of the following is the acceleration at Q?

- A. $0.5a$
- B. a
- C. $2a$
- D. $4a$

19. The gravitational field strength at the surface of a certain planet is g . Which of the following is the gravitational field strength at the surface of a planet with twice the radius and twice the mass?

- A. $\frac{g}{2}$
- B. g
- C. $2g$
- D. $4g$

8. A car on a road follows a horizontal circular path at constant speed. Which of the following correctly identifies the origin and the direction of the net force on the car?

	Origin	Direction
A.	car engine	toward centre of circle
B.	car engine	away from centre of circle
C.	friction between car tyres and road	away from centre of circle
D.	friction between car tyres and road	toward centre of circle

19. The magnitude of the gravitational field strength at the surface of a planet of mass M and radius R is g . What is the magnitude of the gravitational field strength at the surface of a planet of mass $2M$ and radius $2R$?

A. $\frac{g}{4}$

B. $\frac{g}{2}$

C. g

D. $2g$